ANANYA SRIDHAR

Robotics & Computer Vision Engineer — Autonomous Systems, Simulation & Edge CV

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GitHub: github.com/darayavahoush · Portfolio: darayavahoush.github.io · LinkedIn: linkedin.com/in/ananya-sridhar-ai · Vaaksudhi Live · Vaakify Live

Languages: English, Hindi, Tamil

PROFESSIONAL SUMMARY

Robotics and Computer Vision Engineer with hands-on production expertise in physics-backed 3D simulation (MuJoCo), deep reinforcement learning (PPO, Q-Learning), multimodal sensor fusion, and real-time edge perception (OpenCV, DINOv2). Proven track record developing autonomous navigation agents, sub-50ms CPU vision microservices, and interactive 3D robot kinematics simulators. Experienced in building low-latency FastAPI/WebSocket backends and deploying containerized models.

TECHNICAL SKILLS

- **Robotics & Simulation:** MuJoCo 3D Physics Simulation, Gymnasium, Multimodal Sensor Fusion (Lidar, IMU, Thermal), Motion Planning, Forward & Inverse Kinematics, Three.js, React Three Fiber
- **Computer Vision:** OpenCV, MediaPipe Tasks, Vision Transformers (DINOv2), Object Detection & Tracking, Background Subtraction, Edge AI, Feature Matching, Image Preprocessing
- **Deep Learning & Frameworks:** PyTorch, Deep Reinforcement Learning (PPO, Q-Learning), MCTS, A* Pathfinding, Scikit-Learn
- **Systems, Cloud & Backend:** Python, C++, C, SQL, FastAPI, REST APIs, Microservices, WebSockets, Docker, Azure Cloud, Git, Linux, Uvicorn

PROFESSIONAL EXPERIENCE

AI/ML ENGINEER INTERN (COMPUTER VISION & SYSTEMS) | MANAS LEARNING

Mar 2026 – Present | Remote / Mumbai, India

- **Real-Time Visual Perception:** Integrated MediaPipe Tasks Vision for real-time facial landmark tracking and oral-motor movement verification during clinical speech therapy exercises.
- **Adaptive Reinforcement Learning:** Developed a Q-learning difficulty controller dynamically tuning game environments and visual exercises based on user accuracy and reaction times.
- **Low-Latency Streaming Architecture:** Built a modular FastAPI and WebSocket backend streaming sub-50ms perceptual feedback to React client applications.
- **Platform Delivery & Security:** Supported full-stack deployment across vaaksudhi.manaslearning.com and vaakify.manaslearning.com; implemented COPPA compliance and secure authentication.
- **Performance Optimization:** Containerized services using Docker and tuned video processing pipelines for seamless browser execution across low-bandwidth environments.

AI/ML ENGINEER INTERN (CLOUD & DATA PIPELINES) | NIVESH AI

Jun 2026 – Present | Mumbai, India

- **Scalable Data Ingestion Infrastructure:** Engineered cloud data pipelines on Azure handling streaming audio feeds and document ingestion for 10,000+ mutual fund distributors (MFDs).
- **Signal Processing & Noise Filtering:** Implemented digital signal preprocessing (`noisereduce`) and spectral filtering pipelines to isolate audio signals in noisy environments.
- **Automated Workflow Impact:** Slashed advisor documentation turnaround time by 80%, automating manual note-taking and compliance tracking across tens of thousands of client interactions.
- **Cloud Reliability & Monitoring:** Containerized backend services with Docker on Azure, set up asynchronous task workers, and implemented automated API health checks and structured logging.

FEATURED PROJECTS

NIRVANA (FIREBOT) — AUTONOMOUS FIREFIGHTING ROBOT & MUJOCO SIMULATION

Live Render Console · GitHub · Docs

Python, MuJoCo 3D Physics, PyTorch, Gymnasium, FastAPI, WebSockets, Three.js

- Built an autonomous firefighting robot operating inside a MuJoCo 3D physics simulator across multi-room environments with volumetric particle fire dynamics.
- Implemented deep reinforcement learning with multimodal sensor fusion (Lidar, IMU, thermal) for autonomous obstacle avoidance and fire source localization in <45s.
- Deployed a real-time web telemetry console on Render streaming robot state, sensor sweeps, and event logs via FastAPI and WebSockets.

TRIPWIRE — REAL-TIME EDGE COMPUTER VISION VEHICLE COUNTER

Live Demo · GitHub

Python, OpenCV, FastAPI, React, Vite, WebSockets, Docker

- Engineered an edge-ready, CPU-only vehicle detection and tracking pipeline from live CCTV feeds with sub-50ms latency.
- Implemented Gaussian background subtraction, centroid tracking, and directional virtual tripwire algorithms on commodity CPUs without requiring GPUs.
- Shipped a low-latency monitoring dashboard streaming processed video over MJPEG and traffic crossing counts via WebSockets.

GECO-ENHANCED — UNSUPERVISED VISUAL SEMANTIC ALIGNMENT

GitHub Repository

PyTorch, DINOv2 (ViT), Python Optimal Transport, FastAPI, React, Docker

- Built an unsupervised visual correspondence pipeline matching keypoints across different object instances under large viewpoint and appearance shifts.
- Utilized Vision Transformer embeddings (DINOv2) and optimal transport algorithms for zero-shot cross-instance alignment without labeled training data.
- Built an interactive web tool for real-time feature visualization and semantic probe correspondence.

3-DOF ROBOTIC MANIPULATOR SIMULATOR — KINEMATICS & OPTIMIZATION

Live Demo · GitHub

Python, FastAPI, NumPy, SciPy, Three.js, React, Vite, Recharts

- Developed an interactive 3D simulation for 3-DOF planar and spatial robotic arms (RRR and PPP configurations).
- Implemented forward and inverse kinematics solvers using numerical optimization for precision end-effector trajectory tracking.
- Visualized reachable workspace boundaries, manipulability ellipsoids, and joint torque limits in real-time Three.js.

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science & Engineering

SASTRA Deemed University, India

Expected Graduation: 2027

CERTIFICATIONS & RECOGNITIONS

- **SWAYAM – IIT Madras:** Comprehensive Electronics and Embedded Systems *(In-Person Practical Program)*
- **IIoT Certification:** Ministry of Heavy Industries (MHI) Program *(Project-Based Evaluation)*